

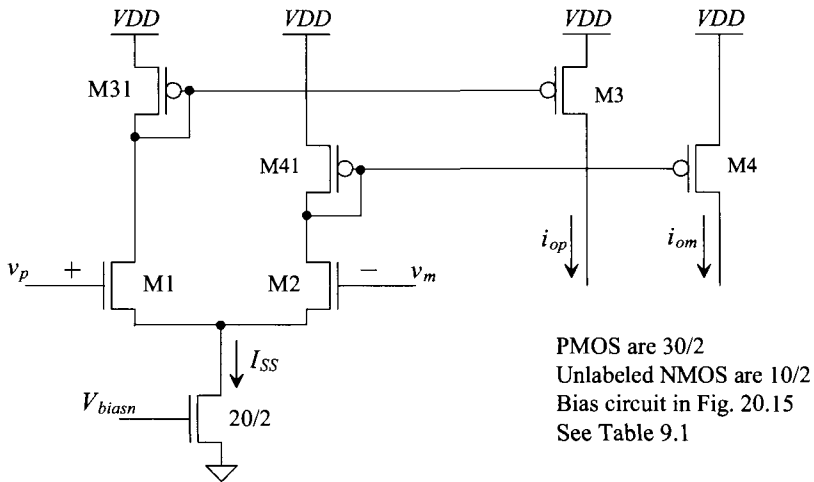


Figure 27.3 Preamplification stage of comparator.

Let's begin by assuming that i_{op} is much larger than i_{om} so that M5 and M7 are on and M6 and M8 are off. We also assume that $\beta_5 = \beta_8 = \beta_A$ and $\beta_6 = \beta_7 = \beta_B$. Under these circumstances, v_{om} is approximately 0 V and v_{op} is

$$v_{op} = \sqrt{\frac{2i_{op}}{\beta_A}} + V_{THN} \quad (27.2)$$

If we start to increase i_{om} and decrease i_{op} , switching starts to take place when the gate-source voltage of M8 is equal to V_{THN} . As we increase M8's V_{GS} beyond V_{THN} (by further increasing i_{om} with the corresponding decrease in i_{op}), M6 starts to take current away from M5. This decreases the drain-source voltage of M5/M6 and thus turns M7 off.

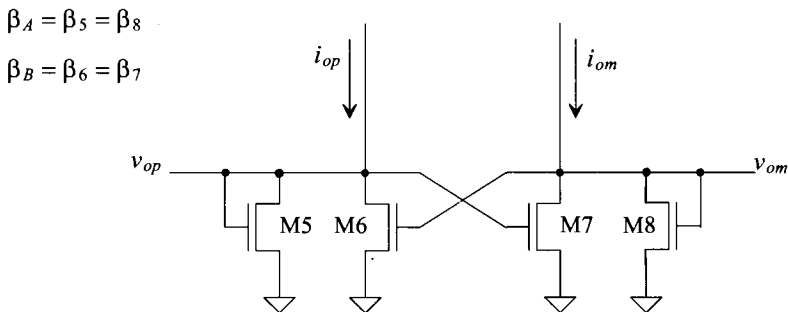


Figure 27.4 Positive feedback decision circuit.

When M8 is just about to turn on (M8's V_{GS} is approaching V_{THN} but the drain currents of M8 and M6 are still zero), the current flowing in M7 is

$$i_{om} = \frac{\beta_B}{2}(v_{op} - V_{THN})^2 \quad (27.3)$$

and the current flowing in M5 is

$$i_{op} = \frac{\beta_A}{2}(v_{op} - V_{THN})^2 \quad (27.4)$$

Noting that the current in M7 (at the switching point) mirrors the current in M5, we can write

$$i_{op} = \frac{\beta_A}{\beta_B} \cdot i_{om} \quad (27.5)$$

If $\beta_A = \beta_B$, then switching takes place when the currents, i_{op} and i_{om} , are equal. Unequal β s cause the comparator to exhibit hysteresis. Relating these equations to Eq. (27.1) yields the switching point voltages (review Ch. 18), or

$$V_{SPH} = v_p - v_m = \frac{I_{SS}}{g_m} \cdot \frac{\frac{\beta_B}{\beta_A} - 1}{\frac{\beta_B}{\beta_A} + 1} \text{ for } \beta_B \geq \beta_A \quad (27.6)$$

and

$$V_{SPL} = -V_{SPH} \quad (27.7)$$

Consider the following example.

Example 27.1

For the circuit shown in Fig. 27.5, estimate and simulate the switching point voltages for two designs: (1) $W_5 = W_6 = W_7 = W_8 = 10$ with $L = 1$ and (2) $W_5 = W_8 = 10$ and $W_6 = W_7 = 12$ with $L = 1$.

For the first case (using the data in Table 9.1)

$$\beta_A = \beta_B = 120 \frac{\mu A}{V^2} \cdot \frac{10}{1} = 1.2 \text{ mA/V}^2$$

so that, as seen in Eq. (27.6) $V_{SPH} = V_{SPL} = 0$. In other words, the comparator does not exhibit hysteresis. Simulation results are shown in Fig. 27.6. The input v_m is set to 2.5 V, while the v_p input is swept from 2.48 to 2.52 V. Note that the amplitudes of v_{op} and v_{om} are limited.

For the second case

$$\beta_A = 120 \frac{\mu A}{V^2} \cdot \frac{10}{1} \text{ and } \beta_B = 120 \frac{\mu A}{V^2} \cdot \frac{12}{1}$$

or $\beta_B = 1.2\beta_A$. Using Eqs. (27.6) and (27.7) with $I_{SS} = 40 \mu A$ and $g_m = 150 \mu A/V$ (again, see Table 9.1), we get

$$V_{SPH} = -V_{SPL} = \frac{I_{SS}}{g_m} \cdot \frac{\frac{\beta_B}{\beta_A} - 1}{\frac{\beta_B}{\beta_A} + 1} = \frac{40}{150} \cdot \frac{1.2 - 1}{1.2 + 1} = 24 \text{ mV}$$

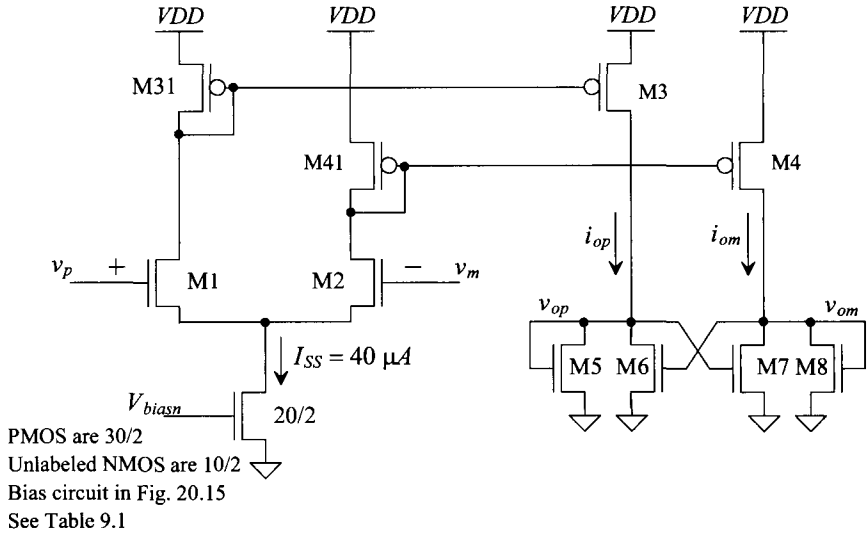
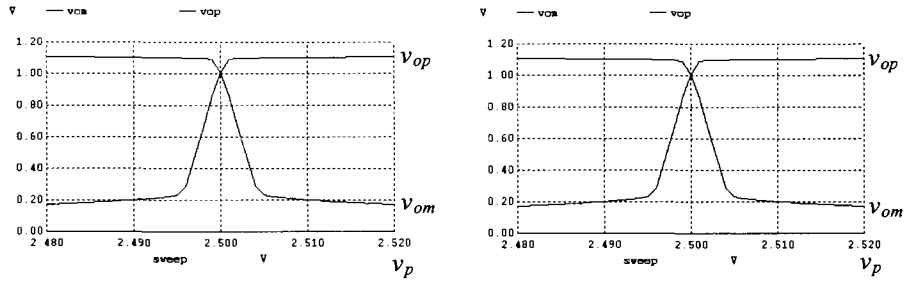


Figure 27.5 Schematic of the pre-amp and decision circuit used in Ex. 27.1.



(a) Minus input held at 2.5 V while the positive input is swept from 2.4 to 2.6 V.
 (b) Minus input held at 2.5 V while the positive input is swept from 2.52 to 2.4.

Figure 27.6 The outputs of the decision circuit in Fig. 27.5 without hysteresis.

The simulation results are shown in Fig. 27.7. Figure 27.7a shows a sweep of v_p from 2.4 to 2.6 V, with v_m held at 2.5 V. Since V_{SPH} is 24 mV, the decision circuit switches states when v_p is 24 mV above v_m (i.e., when $v_p = 2.524$ V). The case of v_p being swept from 2.6 to 2.4 V is shown in part (b) of the figure. Switching occurs in this situation when v_p is approximately 2.476 V, or 24 mV, less than 2.5 V. ■

Output Buffer

The final component in our comparator design is the output buffer or post-amplifier. The main purpose of the output buffer is to convert the output of the decision circuit into a logic signal (i.e., 0 or VDD). The output buffer should accept a differential input signal

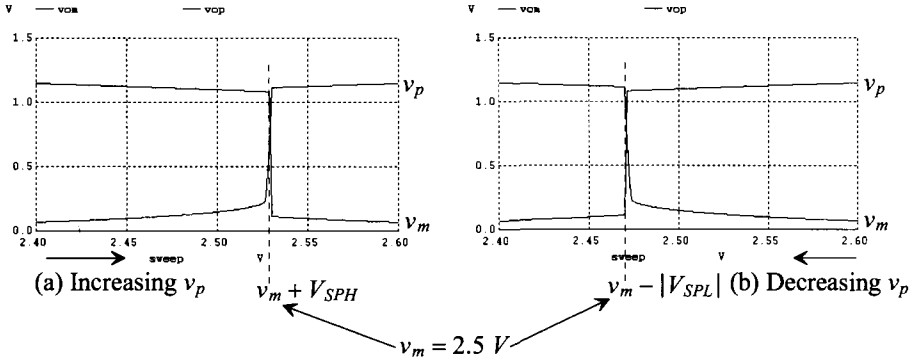


Figure 27.7 Simulating the pre-amp and decision circuit with hysteresis.

and not have slew-rate limitations. We might try to use a NAND SR latch as seen back in Fig. 16.35. However, with process shifts we might run into the situation where the switching points of the gates aren't centered in the middle of the limited swing of the decision circuit's outputs. For a general comparator design, we'll use a diff-amp to help regenerate the digital output signals. For a simple design for the output buffer, we can use the self-biased diff-amp seen back in Fig. 18.17. To move the decision circuit's output swing into the common-mode range of the diff-amp, we can add a gate-drain connected device as seen in Fig. 27.8. An inverter was added on the output of the amplifier as an additional gain stage and to isolate any load capacitance from the self-biasing differential amplifier. This comparator works very well. However, the current in the self-biased stage can be (relatively) huge. This can be a problem if power is a concern (as it is in the Flash ADC discussed in Ch. 29).

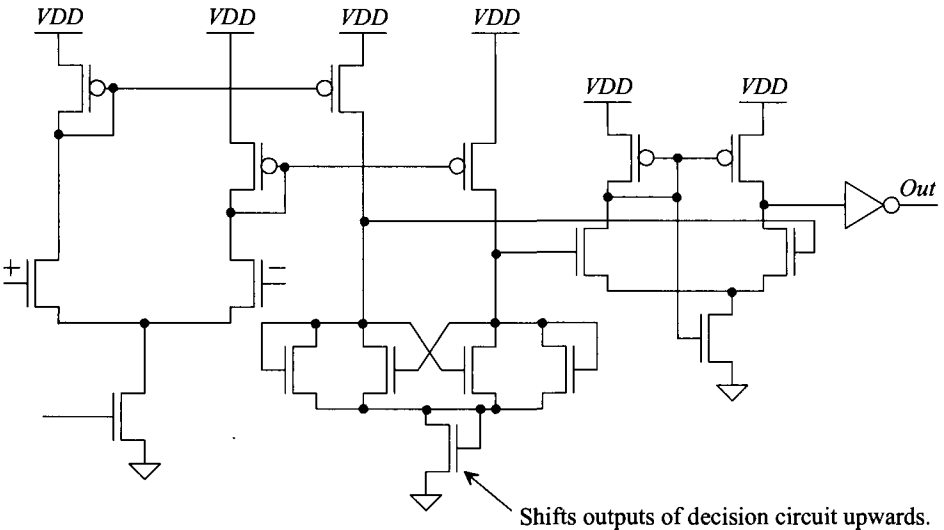


Figure 27.8 Using a self-biased diff-amp for the output buffer.

Figure 27.9 shows a comparator design with both a PMOS and NMOS diff-amp feeding the decision circuit (for input common-mode range beyond the power supply rails). The output buffer is a PMOS diff-amp driving an inverter. While this op-amp won't be as fast as the op-amp seen in Fig. 27.8, it is a good general purpose design. The current drawn from V_{DD} will be considerably less than the design in Fig. 27.8. Note that the output currents of the PMOS diff-amp, on the input of the comparator, are added to the NMOS (current) outputs to ensure that current is always sourced to the decision circuit. We might be tempted to connect the PMOS diff-amp directly to the decision circuit. However, this would result in a larger minimum input common-mode voltage.

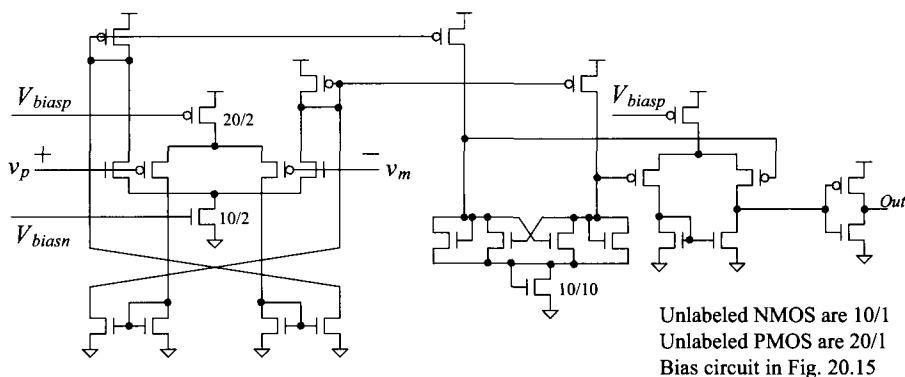


Figure 27.9 General-purpose comparator with rail-to-rail input common-mode range.

27.1.1 Characterizing the Comparator

Comparator DC Performance

Figure 27.10 shows the DC simulated performance of the comparator in Fig. 27.9. The positive comparator input is swept from 0 to V_{DD} ($= 5\text{ V}$) while the negative input is stepped for each simulation from 0 to 5 in 500 mV increments (to show wide input common-mode range). Also seen in this figure is the comparator's current draw from V_{DD} . Included in this current is the current supplied to the bias circuit.

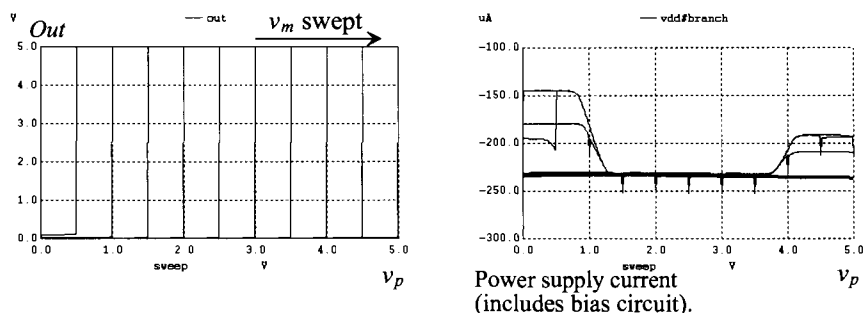


Figure 27.10 DC performance of the comparator in Fig. 27.9

If we take the derivative of these transfer curves, the gain of the comparator and thus the smallest difference that can be discriminated between v_p and v_m becomes known. We have to be careful here with the step size used in the DC simulation. If, for example, we were to use a step size of 1 mV, then the maximum gain we get from the simulation is 1,000. Figure 27.11 shows the expanded view of the comparator with the minus input held at 2.5 V while the positive input is swept from 2.499 to 2.501 V. The offset seen in the figure is a systematic offset (as discussed earlier). The gain of the comparator is, roughly, 175,000. Notice that the comparator was designed without hysteresis. However, in a practical comparator mismatch in the decision circuit results in hysteresis (and so if hysteresis isn't desired, the layout of the decision circuit is critical and may require common-centroid techniques).

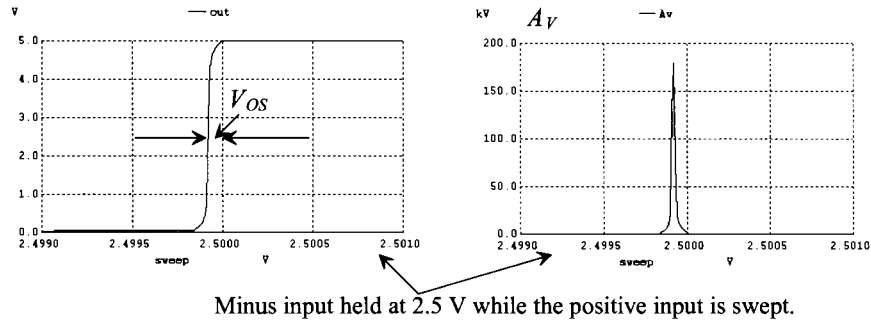


Figure 27.11 The gain of the comparator in Fig. 27.9

Transient Response

The transient response of a comparator can be significantly more difficult to characterize than the DC characteristics. Let's begin by considering $v_m = 2.5$ V DC, with the v_p input to the comparator a 10 ns wide pulse and an amplitude varying from 2.45 to 2.55 V. This is termed a narrow pulse with a 50 mV overdrive; we are driving the + input of the comparator 50 mV over the negative input. The simulation results for these inputs are shown in Fig. 27.12. If the pulse amplitude or width is reduced much beyond this, the comparator does not make a full transition. Notice how we only show simulation data after 120 ns. This was done to ensure the bias circuit had time to start-up and stabilize before changing the inputs of the comparator.

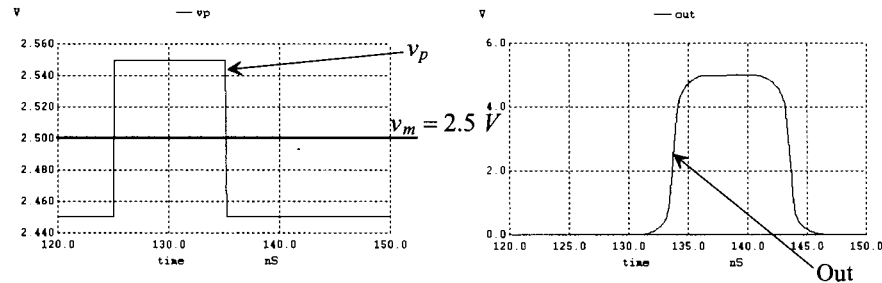


Figure 27.12 Transient response of the comparator in Fig. 27.9.

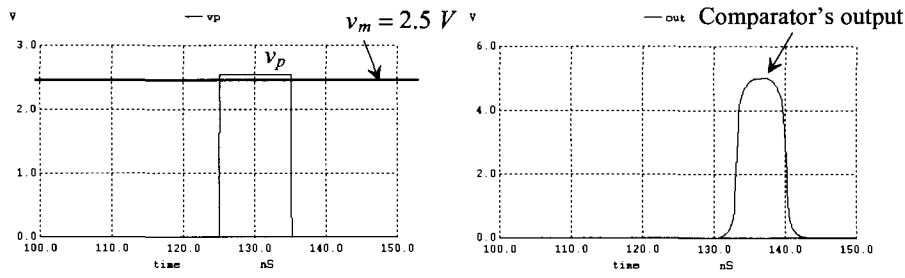


Figure 27.13 Transient response of the comparator in Fig. 27.9 with one side of the pre-amp initially shut off.

Although these results are interesting, they are not practically useful in some situations. In few cases will the comparator discriminate between signals similar to what is seen in Fig. 27.12. A better indication of comparator performance is to apply a signal that starts at a voltage where one side of the differential amplifier is cut off and to finish at a voltage slightly larger than the reference voltage (2.5 V in the above example), Fig. 27.13. In applying a 0 to 2.6 V pulse to the v_p input, the left side of the diff-amp is initially off and then on. The internal comparator nodes must be charged over a wider voltage range with a 0–2.6 V input when compared to the 2.45–2.55 V example of Fig. 27.12. In order to keep one side of the pre-amp from turning off, the circuit of Fig. 27.14 can be used. The added MOSFET ensures that the voltage between the drains of M31 and M41 don't deviate too much. This configuration is sometimes referred to as a clamped input stage. Note that the size of the added MOSFET should be as small as possible to avoid unneeded loading in the diff-amp (which slows down the comparator's performance).

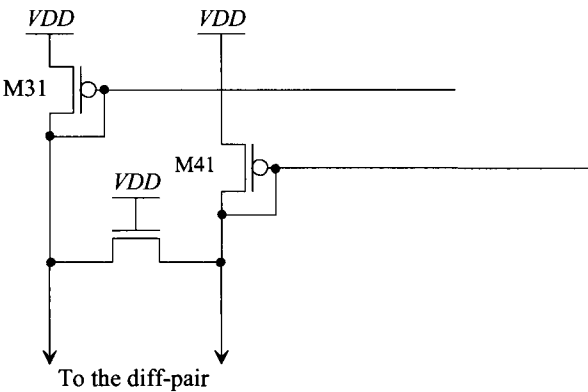


Figure 27.14 Adding a balancing resistor to the input stage of a comparator. See Fig. 27.5